# NIRAJ BHATTA

**AI Engineer | Computer Vision · NLP · Machine Learning**

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## PROFESSIONAL SUMMARY

Computer Engineering undergraduate (3.92 GPA in latest semester, expected graduation 2027) focused on building applied AI systems across computer vision, NLP and retrieval-augmented generation. Built a Nepali number plate recognition model using CNN and LSTM, a YouTube video chatbot using LangChain and RAG, and an IoT smart parking system that won the intra-college IoT competition. Combines Python and full stack development skills with strong leadership and communication experience from organizing college-level technical events. Seeking an AI Engineer role to design, train and deploy practical machine learning solutions.

## TECHNICAL SKILLS

**AI / Machine Learning:** Machine Learning, Deep Learning (CNN, LSTM), Computer Vision, Natural Language Processing, Retrieval-Augmented Generation (RAG), LangChain, F AST-API, OCR (EasyOCR, Tesseract, PaddleOCR), Prompt Engineering

**Programming Languages:** Python, JavaScript, PHP, SQL

**Backend and APIs:** Django, Node.js, Express, RESTful API Development, MySQL, MongoDB

**Frontend:** React.js, HTML, CSS

**IoT and Embedded:** Arduino, Ultrasonic Sensors, IR Sensors, I2C LCD, Hardware Troubleshooting

**Tools:** Git, GitHub

## RESEARCH EXPERIENCE

**IoT Smart Parking Management with Nepali ANPR and Hazard Monitoring** | Capstone Research Project, NCIT | View Paper

- Co-authored a research paper on the system with a team of three, published at ICICSET 2026.
- Developed a tiered Nepali number plate recognition pipeline: YOLOv8s plate localization (mAP@0.50 of about 95%), 34-class CNN character recognition and geometric reconstruction of plate text.
- Built the IoT stack with multi-board ESP32 entry and exit controllers over MQTT, Firebase Realtime Database, RFID access, servo gates, and DHT11, MQ2 and flame sensors for hazard monitoring.
- Built a React and Vite dashboard with slot reservation and auto-expiry logic.
- Evaluated the models with precision-recall curves and confusion matrices in Google Colab, and wrote the paper in LaTeX.

## PROJECTS

**Automatic Number Plate Recognition for Nepali Plates** | Python, CNN, LSTM, Computer Vision | GitHub

- Designed a CNN and LSTM based model to detect and recognize Nepali number plates under real Nepali environmental conditions.
- Integrated the recognition model into an IoT smart parking system with hazard monitoring as part of the final year capstone project.

**YouTube Video Chatbot (RAG)** | Python, LangChain, RAG, NLP | GitHub

- Built a retrieval-augmented generation chatbot that answers questions about YouTube video content using LangChain.
- Published the source code with documentation on GitHub and wrote a blog post explaining the approach.

**Smart Parking System** | Arduino, Ultrasonic Sensor, IR Sensor, I2C LCD | View Image

- Developed an automated parking solution that detects vehicles with sensors and displays available slots in real time.
- Won first place in the Intra-College IoT Competition for this system.

**Document Verification System** | PHP, JavaScript, MySQL, EasyOCR | GitHub

- Built a document verification system with OCR-based text extraction and fraud detection for government documents as Core Developer at Hackathon Orbit 2.0.

- Delivered the working system collaboratively with a multidisciplinary team under strict deadlines.

**Deltaversity** | Node.js | GitHub

- Designed and developed a centralized management platform for a UK-based educational institute covering courses, contacts, blogs, events and website content administration.

**Banking System and Student Information System** | Django, PHP, MySQL | GitHub

- Developed secure web applications with user authentication, transaction handling, attendance and grade management, and admin and user dashboards.

## EDUCATION

**Bachelor of Engineering in Computer Engineering**

Nepal College of Information Technology (NCIT), Pokhara University | Expected Graduation: 2027

- GPA: 3.92 in the last semester
- Relevant Coursework: Artificial Intelligence, Machine Learning, Natural Language Processing, Computer Graphics

## CERTIFICATIONS

- Django Development (30 Hours)
- Hardware Fellowship (7 Days): electronics, sensors and hardware systems
- Prompt Engineering Basics

## LEADERSHIP AND ACHIEVEMENTS

- **Winner, Intra-College IoT Competition, NCIT:** First place with the Smart Parking System.
- **Core Developer, Hackathon Orbit 2.0:** Built a document verification system using OCR and web technologies.
- **Program Coordinator, IEEE NCIT Student Branch:** Coordinated technical workshops, seminars and networking sessions.
- **Organizer, NCIT TechFest:** Planned and coordinated the Hackathon, Tech Talk sessions and other key activities.
- **Lead Organizer, Host and Designer, CITC (Computer and IT Club):** Led event operations, hosted sessions and designed promotional materials.
- **Sponsorship Team Member, TEDxNCIT:** Secured sponsorships and partnerships for event funding.
- **Secretary, Leo Club of Kathmandu Bihani:** Managed communications and supported community service initiatives.
- **Winner, NCIT Football Championship:** Inter-college champion.

## LANGUAGES

Nepali (Native) | English (Professional Working Proficiency) | Hindi (Conversational)

## REFERENCES

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